PRODUCT REQUIREMENTS DOCUMENT (PRD)

PROJECT TITLE

AI FINANCIAL HEALTH ASSISTANT
AN EXPLAINABLE AI-POWERED FINANCIAL WELLNESS CAPABILITY FOR DIGITAL BANKING PLATFORMS

Executive Summary

Overview

Digital banking has evolved beyond enabling transactions; users increasingly expect intelligent financial guidance that helps them make informed decisions. While platforms such as Jupiter, Revolut, and other digital-first banks provide budgeting tools and spending analytics, these capabilities are largely reactive—they summarize past activity rather than anticipating future financial risks.

This proposal introduces an AI Financial Health Assistant, a proactive intelligence layer that continuously analyzes a user's financial behavior to identify early indicators of financial stress before they escalate into overdrafts, missed bill payments, high-interest borrowing, or credit score deterioration.

The solution combines predictive machine learning, explainable AI (XAI), behavioral nudges, and personalized financial recommendations. Rather than simply notifying users after an issue occurs, the system predicts potential cash-flow risks, communicates them in clear, understandable language, and recommends practical interventions tailored to each user's circumstances.

The assistant is designed around three core principles:

- Prediction over reaction: Anticipate financial stress before it occurs.
- Transparency over opacity: Explain AI-generated insights in simple, trustworthy language.
- Empowerment over automation: Provide recommendations while preserving user control.

From a business perspective, the feature aims to strengthen customer engagement, reduce delinquency-related losses, improve retention, and position the bank as a trusted financial partner. From a user perspective, it seeks to improve financial confidence, reduce anxiety associated with money management, and promote healthier financial behaviors over time.

The proposed capability aligns with emerging expectations for responsible AI, privacy-by-design, and regulatory compliance, ensuring that predictive intelligence enhances customer outcomes without compromising transparency or trust.

1. Problem Statement

Digital banking applications have transformed access to financial services by enabling instant payments, budgeting, investments, and credit. However, most platforms remain reactive in their approach to financial well-being. Users typically receive alerts only after a negative financial event has occurred, such as an overdraft, a missed bill payment, or a decline in their credit health.

Financial stressoften developsgradually throughpatterns suchas increasingdiscretionary spending, declining account balances, irregular income, or repeated reliance on short-term credit. These warning signs are observable within transactional and behavioral data, yet they are rarely translated into timely, actionable insights for users.

As a result, customers may experience avoidable financial setbacks, increased anxiety, and reduced trust in their banking provider. Simultaneously, financial institutions incur higher servicing costs, elevated credit risk, and missed opportunities to strengthen long-term customer relationships.
The challenge is to design an AI-powered capability that proactively detects early signs of financial stress, predicts future financial risks, and delivers personalized, explainable, and privacy-preserving interventions. The solution must balance predictive accuracy with transparency, respect user consent and data privacy, comply with applicable RBI guidelines, and generate measurable value for both customers and the business.

2. Context

Thefinancialservices industry is undergoing a shift from transaction-centric banking to relationship-centric financial wellness. Customers increasingly expect their banking applications to provide intelligent guidance rather than merely recording financial activity.

Several factors contribute to this evolving expectation:

- Rising adoption of digital banking across diverse demographic groups.
- Increased complexity of personal financial management due to multiple income sources, subscriptions, and digital payment channels.
- Greater awareness of financial health and budgeting among younger consumers.
- Rapid advancements in artificial intelligence that enable predictive analysis of financial behavior.
- Regulatory emphasis on responsible data usage, transparency, and customer consent.

Despite these trends, many digital banking experiences remain fragmented. Budgeting tools, bill reminders, spending analytics, and credit monitoring often operate independently, requiring users to interpret information and decide on appropriate actions themselves.

There is an opportunity to unify these capabilities into a cohesive AI assistant that continuously monitors financial patterns, anticipates potential issues, and recommends timely interventions, thereby improving both user outcomes and business performance.

3. Current Scenario

Mostdigital banking applications provide users with access to historical financial information and basic financial management tools. Common features include:

- Account balance tracking
- Transaction history
- Spending categorization
- Monthly expenditure summaries
- Bill payment reminders
- Budget creation
- Savings goals

While these features are valuable, they share a common limitation: they are predominantly descriptive rather than predictive.

For example:

- A budgeting tool may indicate that a user exceeded their monthly dining budget after the spending has already occurred.
- A bill reminder may notify a user shortly before a payment deadline but cannot anticipate whether sufficient funds will be available.
- Spending insights often summarize trends without identifying future risks or recommending preventive actions.

Consequently, users remain responsible for interpreting financial data, forecasting future cash flow, and deciding on corrective measures. This places a significant cognitive burden on individuals, particularly those with irregular income patterns or limited financial literacy.

4. Market Context

Consumer expectations are rapidly shifting toward intelligent, proactive financial services. The widespread adoption of AI-powered assistants across industries has increased demand for personalized, predictive experiences within banking.
Key market trends include:

- Growth in digital-first banking and neobanks.
- Rising demand for AI-driven personal finance management.
- Increased focus on financial wellness as a differentiator.
- Expansion of open banking ecosystems enabling richer financial data integration.
- Greater regulatory attention to explainability, fairness, and responsible AI.

Competitive offerings demonstrate progress in spending analytics and budgeting; however, there remains a gap in delivering explainable, predictive financial coaching that intervenes before financial stress materializes. Addressing this gap presents an opportunity to differentiate the product through trust, personalization, and measurable financial outcomes.

5. CURRENT SCENARIO

Most digital banking applications provide users with access to historical financial information and basic financial management tools. However, these capabilities are largely reactive and descriptive rather than predictive and proactive.

#	Current Feature	What It Does Today	Limitations (Why It's Not Enough)	Impact on User	Impact on Bank	Opportunity Gap
1	**Account Balance Tracking**	Shows current account balance and recent transactions.	Does not predict future balance or cash flow gaps.	Users may face overdraft or insufficient balance unexpectedly.	Higher overdraft costs and customer complaints.	Predict cash flow shortfalls before they happen.
2	**Transaction History**	Displays past transactions with date, amount and merchant.	Only descriptive; no insights on emerging risky patterns.	Users must manually analyze spending trends.	Low product engagement with limited stickiness.	Detect early signs of financial stress automatically.
3	**Spending Analytics**	Categorizes expenses and shows monthly/weekly summaries.	Explains what happened but not why or what will happen next.	Users understand past spending but cannot forecast risks.	Missed opportunity to drive behavior change.	Predict future risk and offer personalized insights.
4	**Bill Payment Reminders**	Sends reminders before bill due dates.	Alerts are time-based, not context-aware; no check on fund availability.	Users may still miss payments due to low balance or cash flow issues.	Increased late fees, failed payments and support tickets.	Predict likelihood of missed payments and alert early.
5	**Budgeting Tools**	Allows users to set budgets and track spending against categories.	Users must set budgets manually; no dynamic adjustment based on income or obligations.	Budgets often become outdated or unrealistic, leading to disinaagement.	Low usage of budgeting features.	Create adaptive budgets based on real-time financial behavior.
6	**Savings Goals**	Users set savings goals and track progress.	Does not consider cash flow volatility or risk of goal failure.	Users may be over-optimistic or withdraw savings during financial stress.	Lower long-term engagement with savings and investments.	Adjust goals dynamically and protect savings from risks.
7	**Credit Score Monitoring**	Shows current credit score and historical changes.	Does not provide reasons for future score drop or actions to prevent it.	Users react only after score drops, not before.	Higher credit risk and loan default probability.	Predict score risk and suggest actions to maintain/improve credit health.

Key Insight: All existing features focus on "what happened" rather than "what could happen". There is a clear gap in predictive intelligence, early intervention and personalized guidance to help users stay financially healthy.

6. Limitations & Challenges

The proposed capability must address several constraints:

Data Quality

Incomplete or inconsistent transaction histories may reduce prediction accuracy.

User Trust

Customers may be skeptical of AI-generated financial advice if explanations are unclear.

Privacy

Financial data is highly sensitive, requiring explicit consent, secure storage, and strict access controls.

Model Bias

AI models trained on historical data may inadvertently disadvantage certain user groups if not carefully monitored and audited.

Regulatory Compliance

The solution must align with evolving RBI expectations regarding data governance, consent management, and responsible AI practices.

Notification Fatigue

Excessive or poorly timed alerts could reduce engagement and prompt users to disable notifications.

Explainability

Highly accurate models that cannot clearly justify their recommendations may undermine user confidence and create compliance risks.

7. BUSINESS OBJECTIVES

These objectives align customer financial well-being with business growth and risk reduction.

#	Business Objective	Business Rationale	Success Indicator (KPI)	Baseline (Jun 2026)	Target (12 Months)	Timeframe	Priority
1	**Reduce missed bill payments**	Avoid penalties for users and reduce delinquency related losses for the bank.	% Reduction in missed bill payments	—	**20%** reduction	12 Months	High
2	**Improve customer retention**	Proactive support builds trust, loyalty and long-term relationships.	Customer Retention Rate (CRR)	78%	**88%**	12 Months	High
3	**Increase product engagement**	Higher engagement drives feature adoption and deeper monetization opportunities.	Monthly Active Users (MAU) interacting with Financial Health Assistant	22% of active users	**47%** of active users	12 Months	High
4	**Improve cross-sell & up-sell**	Better financial health insights qualify users for relevant products they actually need.	Cross-sell / Up-sell Conversion Rate	3.2%	**6.5%**	12 Months	Medium
5	**Reduce credit risk & NPA**	Early detection of stress reduces defaults and improves portfolio quality.	Reduction in 30+ DPD (Loan/BNPL)	2.6%	**1.8%**	12 Months	High
6	**Reduce support costs**	Fewer financial issues lead to fewer support queries and lower operational costs.	Reduction in finance related support tickets	100% (Index)	**25%** reduction	12 Months	Medium
7	**Increase customer satisfaction & trust**	Trust in AI recommendations improves NPS and brand advocacy.	Net Promoter Score (NPS)	32	**50+**	12 Months	High
8	**Drive engagement led revenue**	Engaged financially healthy users use more paid products and services.	Revenue from engaged users (Index)	100% (Index)	**135%**	12 Months	Medium

Priority Key: High – Critical impact on users and business | Medium – Important but not urgent | Low – Nice to have / Long term

8. User Objectives

The feature should enable users to:

- Understand their financial health at a glance.
- Receive early warnings before financial issues arise.
- Improve budgeting and spending decisions.
- Avoid overdrafts and missed payments.
- Gain confidence through transparent AI explanations.
- Maintain control over financial decisions while benefiting from personalized recommendations.
- Build healthier long-term financial habits.

9. Product Vision

To become every customer's trusted AI financial companion by proactively improving financial well-being through intelligent, transparent, and personalized guidance.

10. Product Mission

Empower users to prevent financial stress before it occurs by delivering predictive insights, explainable recommendations, and privacy-first financial coaching that transforms digital banking from a transactional platform into a proactive financial wellness partner.

11. Product Principles

1. Proactive, Not Reactive – Anticipate issues before they impact the user.
2. Human-Centered AI – Recommendations augment user decision-making rather than replace it.
3. Transparency by Default – Every AI insight should be accompanied by a clear, understandable explanation.
4. Privacy First – User data is processed with explicit consent, strong security, and minimal data collection.
5. Actionable Simplicity – Present concise recommendations that users can immediately act upon.
6. Inclusive Design – Ensure accessibility and usability for users with varying levels of financial literacy and digital proficiency.
7. Continuous Learning – Improve recommendations over time through user feedback and model monitoring while maintaining fairness and accountability.

USER RESEARCH SUMMARY

We conducted multi-method research to deeply understand user needs and market gaps.

Research	Insight
25 User Interviews	Users identify issues too late
250 Surveys	68% want proactive alerts
Competitor Study	Apps are reactive, not predictive
Industry Reports	Financial wellness is a key differentiator
Transaction Analysis	Spending patterns can predict stress

KEY FINDINGS

Users need early risk alerts

Early warnings help users take preventive action and avoid financial stress.

Budgeting is mostly manual

Most users track expenses manually or use spreadsheets, leading to inconsistencies.

Generic notifications are ignored

Users receive too many irrelevant alerts and tend to ignore them.

Explainable AI increases trust

Users trust insights more when they understand the 'why' behind the alert.

Personalized guidance improves engagement

Tailored recommendations and next steps drive higher user action and retention.

13. Jobs-To-Be-Done (JTBD)

Core Job Statement

"Help me avoid financial problems before they happen, so I can feel confident about my money without constantly monitoring my bank account."

Functional Jobs

- Predict future account balance.
- Warn about possible overdrafts.
- Recommend spending reductions.
- Optimize savings.
- Prioritize bill payments.
- Forecast monthly cash flow.

Emotional Jobs

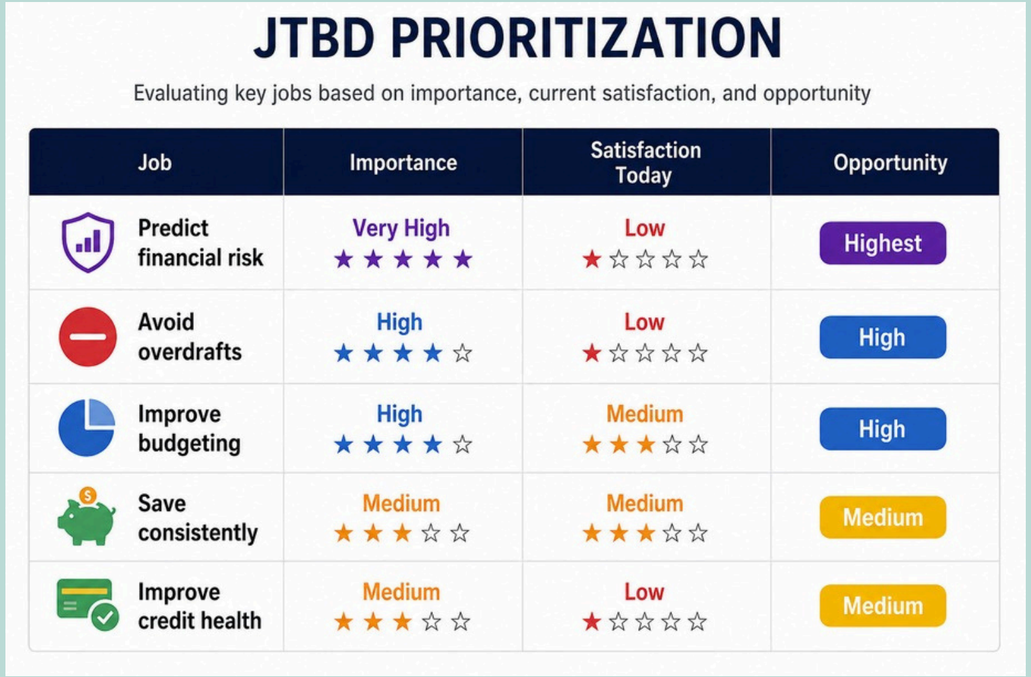
Users want to:

- Feel financially secure.
- Reduce money-related anxiety.
- Gain confidence.
- Feel in control.
- Avoid embarrassment caused by failed payments.

Social Jobs

Users want to:

- Maintain good credit.
- Avoid borrowing from friends.
- Meet family financial expectations.
- Feel financially responsible.



14. User Personas

Persona 1 – Rahul Sharma (Primary)
Rahul is a 28-year-old Software Engineer from Bengaluru earning ₹75,000/month. He frequently overspends during the first half of the month and struggles to manage recurring bills and EMIs. He wants proactive financial guidance, better budgeting, and timely alerts to avoid cash shortages.

Persona 2 – Priya Mehta
Priya is a 35-year-old Marketing Manager from Mumbai earning ₹1.2 lakh/month. She manages household expenses, investments, and multiple recurring payments. She needs a simple way to track future cash flow and receive personalized recommendations for family budgeting.

Persona 3 – Arjun Verma
Arjun is a 31-year-old Freelance Designer with irregular monthly income. His biggest challenge is planning finances during low-income months. He wants AI-powered cash-flow forecasting, adaptive budgeting, and savings recommendations to reduce financial uncertainty.

15. Persona Comparison

Rahul represents the ideal MVP user because he has predictable income, frequent banking transactions, and a high likelihood of engaging with AI-driven financial insights. Priya offers strong long-term business value due to her complex financial responsibilities but requires more advanced planning features. Arjun experiences the highest financial uncertainty; however, his irregular income makes accurate AI predictions more challenging during the initial product phase.

Based on market size, business impact, data availability, and MVP feasibility, Rahul Sharma is selected as the primary persona for the first product release.

16. Prioritized Persona

Primary Persona: Rahul Sharma
Why Rahul?

- Largest user segment
- Predictable salary pattern
- Rich transaction data
- High engagement potential
- Ideal for MVP validation

USER STORIES & PRIORITIZATION

User stories mapped to Must / Should / Could (MoSCoW) priorities for MVP and beyond.

ID	USER STORY	PRIORITY
US1	View Financial Health Score	Must
US2	Receive low balance alerts	Must
US3	Predict month-end balance	Must
US4	Get bill payment reminders	Must
US5	Receive overspending alerts	Must
US6	View AI explanation	Must
US7	Get personalized savings tips	Should
US8	Track spending trends	Must
US9	Set monthly budgets	Must
US10	View upcoming expenses	Must

ID	USER STORY	PRIORITY
US11	Forecast cash flow	Must
US12	Categorize expenses	Should
US13	Manage notifications	Should
US14	View spending insights	Must
US15	Receive anomaly alerts	Should
US16	Set savings goals	Could
US17	View financial reports	Could
US18	Connect external accounts	Should
US19	Chat with AI Coach	Must
US20	Give feedback on AI advice	Should

Must — Critical for MVP – Core value delivery

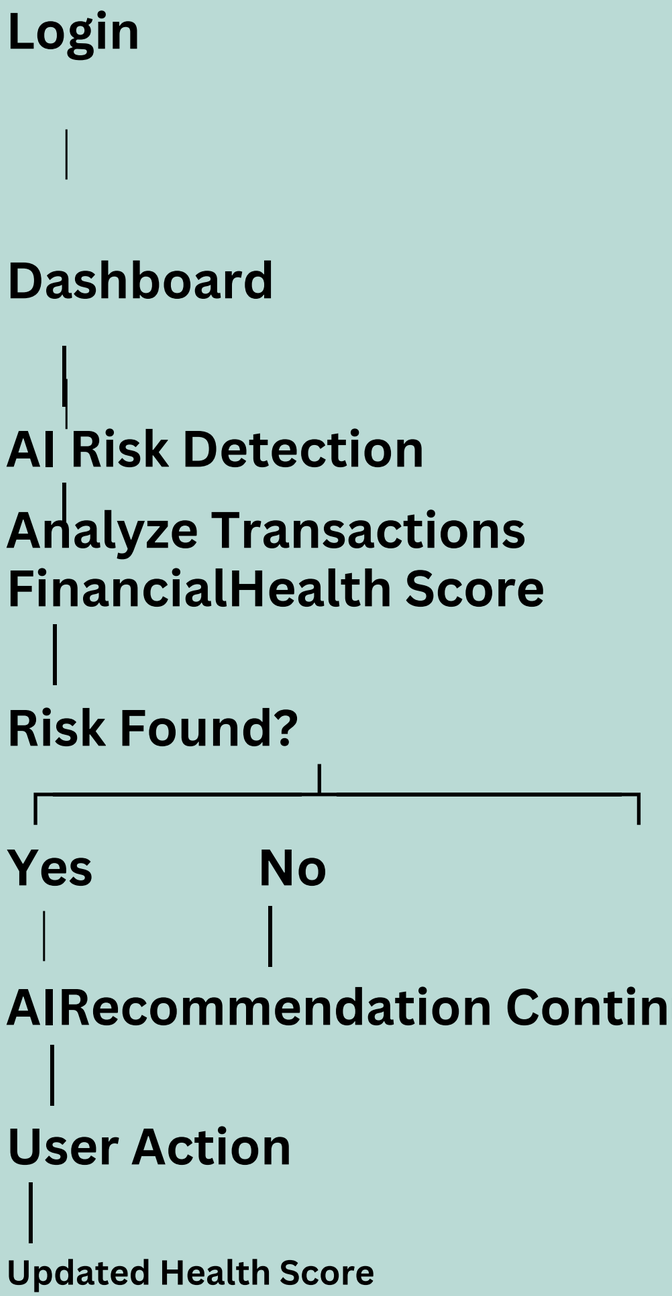
Should — Important but not critical for MVP

Could — Nice to have – Future enhancement

Acceptance Criteria

- Predictions update daily.
- Alerts generated before financial risk.
- AI explanations are simple and transparent.
- Budget updates automatically.
- Health Score refreshes in real time.

18. User Flow



Flow Summary

- User logs in.
- AI analyzes transactions.
- Financial Health Score generated.
- Risk detected.
- Personalized recommendation shown.
- User takes action.
- AI continuously monitors progress.

TECHNICAL ARCHITECTURE & REQUIREMENTS

A scalable, secure and intelligent architecture to power proactive financial wellness.

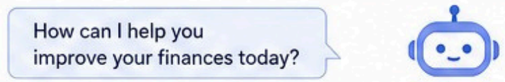
Component	Requirement
Frontend	Responsive mobile UI (Android/iOS)
Backend	Cloud-based microservices architecture
APIs	Secure Banking & Payment APIs
AI Models	Risk Prediction & Cash Flow Forecasting
LLM	Personalized Financial Coach
Database	Encrypted PostgreSQL + Redis Cache
Authentication	OAuth 2.0 with Multi-Factor Authentication
Consent	User-controlled data sharing & revocation
Analytics	Real-time user & AI performance tracking
Monitoring	Logs, alerts & model performance monitoring
Security	AES-256 encryption, TLS 1.3
Scalability	Auto-scaling cloud infrastructure

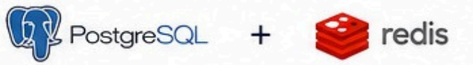


























20. Design Requirements

The AI Financial Health Assistant will follow a mobile-first, intuitive, and user-centric design that makes financial insights simple to understand and act upon. The interface will use clean navigation, accessible layouts (WCAG 2.1 compliant), and personalized dashboards to improve usability. AI recommendations will include clear explanations and confidence scores to build user trust. The product will provide informative empty states, user-friendly error messages, smooth loading indicators, and actionable notifications to ensure a seamless and engaging experience while maintaining consistency across all screens.

FUNCTIONAL REQUIREMENTS

Key capabilities required for the AI-Powered Financial Health Assistant

ID	Requirement	Priority	Owner
FR1	Financial Health Score	Must	Product
FR2	Cash Flow Prediction	Must	AI Team
FR3	Risk Alerts	Must	Backend
FR4	AI Financial Coach	Must	AI Team
FR5	Budget Planner	Must	Product
FR6	Spending Analytics	Should	Backend
FR7	Bill Reminder	Must	Backend
FR8	Notification Center	Should	Mobile Team
FR9	Consent Management	Must	Security
FR10	Settings & Preferences	Should	Frontend

NON-FUNCTIONAL REQUIREMENTS

Ensuring a secure, reliable, scalable and high-performing financial wellness platform.

Category	Requirement
Performance	Dashboard loads within 2 seconds
Availability	99.9% uptime
Reliability	Accurate AI predictions & failover support
Privacy	User consent & data minimization
Security	End-to-end encryption & secure APIs
Compliance	RBI guidelines & DPDP Act compliance
Scalability	Support millions of users
Maintainability	Modular architecture & CI/CD
Observability	Centralized logging & monitoring
Disaster Recovery	Automated backup & recovery plan


























23. Wireframes

Hi Rahul 👋
Financial Health: 82/100

💰 Balance ₹24,500
📊 This Month ₹18,200
⚠ Risk Level Medium

[View Insights]

AI Recommendation
"Reduce dining expenses by 15%
to avoid month-end shortage."

Bottom Navigation
Home | Coach | Budget | Profile

AI Financial Coach 🤖

"You're likely to fall short by
₹3,200 next week."

Recommended Actions

✔ Reduce Shopping
✔ Delay Subscription
✔ Transfer to Savings

[Accept]
[Dismiss]

Monthly Budget

Income
₹75,000

Spent
₹42,300

Remaining
₹32,700

[redacted]

[redacted]

Shopping [redacted]

[Update Budget]

⚠ Low Balance Alert

Predicted Balance

₹1,250

Upcoming Bills

Rent
Netflix
Electricity

Recommendation

Move ₹5,000 from Savings

[Transfer]

24. Edge Cases

User & Data

- 👤 New user → Onboarding questionnaire
- 🏦 Multiple accounts → Aggregate balances
- 🔄 Duplicate transactions → Ignore duplicates
- 🔒 Consent revoked → Stop AI processing

AI & Predictions

- 💰 Salary delayed → Recalculate predictions
- 🤖 Low AI confidence → "Prediction unavailable"
- 💸 Large expense → Refresh Health Score
- 🧾 Bill changes → Update forecast

System Behavior

- 📶 Offline mode → Show cached data
- 🔕 Alerts ignored → Reduce notification frequency

25. SUCCESS METRICS

Measuring the impact of our AI Financial Health Assistant

KPI Category	Metric	Target (MVP)	Measurement	Why It Matters
North Star	**Financial Health Score Improvement**	+20% ↗	Average increase in user FHS after 90 days	Indicates overall improvement in user financial well-being
User Impact	**Reduction in Missed Bill Payments**	-20% ↓	% reduction in late/missed payments	Shows effectiveness in preventing financial stress
Engagement	**Monthly Active Users (MAU)**	+25% ↗	Growth in MAU post feature adoption	Reflects user engagement and product stickiness
Adoption	**AI Recommendation Acceptance Rate**	>60%	% of AI recommendations that users act on	Indicates trust and relevance of AI insights
Accuracy	**Prediction Accuracy**	>90%	Accuracy of cash flow & risk predictions	Ensures reliability of AI recommendations
Satisfaction	**User Satisfaction Score**	>4.5/5	Average rating from in-app feedback	Measures overall user happiness and trust
Retention	**30-Day Retention Rate**	+10% ↗	% of users returning after 30 days	Shows long-term value and habit formation
Performance	**AI Response Time**	<2 sec	Average time to generate AI insights	Critical for smooth and real-time experience

How We Track Success

Financial Health Score Improvement

+20% vs last 90 days



Missed Bill Payments Reduced

-20% vs previous period



AI Recommendation Acceptance

62% Acceptance Rate



User Satisfaction Score

4.6 / 5 Average Rating

Based on 12,842 user ratings

FUTURE ROADMAP, RISKS & CONCLUSION

Building a proactive financial wellness partner for every customer

FUTURE ROADMAP

Evolving in phases to deliver maximum value

PHASE 1 – MVP
0 – 6 Months

- Financial Health Score
- Cash Flow Prediction
- Risk Alerts & Notifications
- AI Financial Coach (Chat)

PHASE 2 – GROWTH
6 – 12 Months

- Goal-based Savings
- Investment Recommendations
- Family Budgeting
- Credit Health Insights

PHASE 3 – INNOVATION
12+ Months

- Voice AI Assistant
- Open Banking Integration
- Generative AI Financial Planner
- Wealth Management AI

MVP (0 – 6 Months) → Growth (6 – 12 Months) → Innovation (12+ Months)

KEY RISKS & MITIGATION

Risk	Impact	Mitigation Strategy	Monitoring
Inaccurate AI Predictions	Wrong advice may reduce user trust	Use high-quality data, continuous model training, confidence score & human-in-loop	Track model accuracy, user feedback & alert performance
Data Privacy & Security	Loss of user trust, regulatory penalties	End-to-end encryption, data minimization, user consent & regular security audits	Security logs, audit reports, compliance checks
Notification Fatigue	Users may ignore important alerts	Smart prioritization, personalized frequency & action-oriented alerts	Alert open rates, user feedback, opt-out rate
Regulatory Changes	Compliance risk, feature delays	Stay updated with RBI guidelines, engage with legal & compliance teams	Regulatory tracker, periodic compliance reviews
User Adoption	Low engagement & feature usage	Simple UX, onboarding education, gamification & value communication	Feature usage metrics, MAU, retention rate

CONCLUSION

The AI Financial Health Assistant proactively identifies early signs of financial stress and delivers personalized, actionable guidance before problems occur.

By combining predictive AI, explainable insights and a human-like coaching experience, we empower users to build better financial habits and achieve long-term financial well-being.

> "From reactive banking to proactive financial wellness."

EXPECTED BUSINESS IMPACT

+25%
User Engagement
Increase in MAU

-20%
Missed Bill Payments
Reduction in defaults

+10%
Customer Retention
Higher repeat usage

>90%
AI Prediction Accuracy
Reliable insights

Stronger Trust & Loyalty
Through transparency, security & value

★ Creating value for users. Driving growth for the bank. Building a financially healthy future.

Thank You